TEXORA · LECTURE SHEET

Quality Systems: ISO 9000, Auditing, QS 9000, ISO 14000, Quality Council and Teamwork

How quality becomes a documented, audited system — the ISO 9000 family and the process approach, quality auditing in its three forms, the QS 9000 automotive system and its successor, environmental management under ISO 14000, and the organisational machinery of quality councils and teamwork.

TE-4108 · QUALITY ASSURANCE AND MANAGEMENT IN TEXTILES · TOPIC 5 · LEVEL 4 TERM 1

1. Quality systems and the ISO 9000 family

A **quality management system (QMS)** is the set of interrelated processes, responsibilities, documents and records by which an organisation directs quality. The **ISO 9000 family** standardises it: **ISO 9000** gives fundamentals and vocabulary; **ISO 9001** states the requirements against which organisations are certified; **ISO 9004** guides sustained success beyond the requirements. ISO 9001 is built on the **process approach** run as PDCA, and on seven quality management principles: customer focus, leadership, engagement of people, process approach, improvement, evidence-based decision making, and relationship management — the ISO restatement of the TQM principles of Topics 1–2.

An ISO 9001 QMS in a textile mill means, concretely: documented processes and responsibilities; control of documents and records; management review at planned intervals; competence and training records; control of production with identified, traceable lots; calibrated measuring equipment (the GSM balance and spectrophotometer have calibration stickers with dates that mean something); control of nonconforming output; and corrective action driven by internal audits and data analysis. Certification is by an accredited third-party body after a documentation review and an on-site audit, with surveillance audits thereafter — the certificate attests that the *system* conforms, which is why a certified mill can still ship a bad lot (Topic 1: a system is necessary, not sufficient) but an uncertified one cannot prove how it works at all.

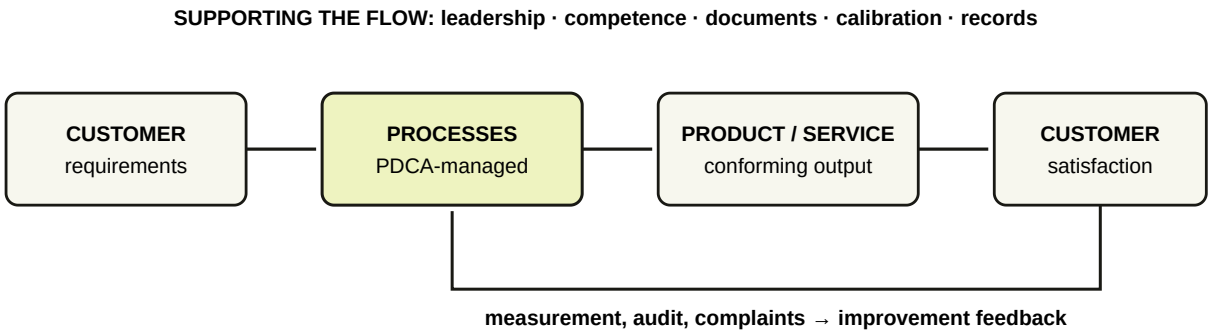


Fig. 1 — The process approach of ISO 9001: requirements in, satisfaction out, and the whole flow wrapped in feedback and support processes.

2. Quality auditing

A **quality audit** is a systematic, independent examination to determine whether activities and results comply with planned arrangements and whether those arrangements achieve objectives. Audits are classified by party:

TYPE	WHO AUDITS WHOM	PURPOSE
First-party (internal)	Organisation audits itself	Find weaknesses before others do; drive corrective action; required by ISO 9001
Second-party	Customer audits supplier	Buyer audits of mills — capability, systems, compliance readiness
Third-party	Independent accredited body	Certification and surveillance (ISO 9001, ISO 14001, social standards of Topic 6)

The audit cycle: plan (scope, criteria, checklist from the standard and the organisation's own documents) → opening meeting → evidence gathering by interview, observation and records (auditors audit *evidence*, not opinions) → findings classified (major nonconformity: a requirement absent or systemically failed; minor: isolated lapse; observation: improvement opportunity) → closing meeting and report → corrective action with root cause (Topic 2's complaint discipline) → follow-up verification. An audit that ends at the report is an expense; the corrective-action loop is where its value sits. Auditor qualities: independence from the area audited, training, and evidence discipline.

3. QS 9000 — the automotive quality system

QS 9000 was the automotive industry's QMS standard, built by the US "Big Three" (Ford, General Motors, Chrysler) on ISO 9001:1994 plus automotive-specific requirements — notably **APQP**

(advanced product quality planning), **PPAP** (production part approval process), **FMEA** and **MSA/SPC** manuals — imposed on their supply chains. QS 9000 certification ended in 2006, superseded by **ISO/TS 16949**, today **IATF 16949**. It belongs in a textile course for two reasons: its tools (FMEA of Topic 3, SPC, structured product planning) diffused into general industry, and technical-textile suppliers *to* the automotive chain (seat fabrics, airbag fabrics, interior textiles) must satisfy its successor standard today — a live example of a customer-specific QMS layered on ISO 9001.

4. ISO 14000 — environmental management

The **ISO 14000 family** applies the same system logic to the environment; its certifiable core is **ISO 14001**, requirements for an **environmental management system (EMS)**: identify environmental aspects and impacts (for a dye house: water use, effluent load, chemical handling, sludge, energy), maintain an environmental policy, comply with legal and other requirements (in Bangladesh, the Department of Environment's ECR rules and ETP discharge standards), set objectives, control operations, monitor, audit, review — PDCA again. ISO 14001 certifies the management system, not a fixed emission level: a certified plant must demonstrate control and improvement, and its legal compliance is audited as part of the system. For export textiles, ISO 14001 commonly sits beside buyer environmental requirements (effluent and chemical programmes of Topic 6).

5. The quality council and teamwork

TQM needs an organisational engine room. The **quality council** — senior managers chaired by the chief executive — owns quality strategically: sets policy and objectives (Topic 2), reviews system performance (audit results, CoQ, complaints, ratings), allocates resources to improvement priorities, and reviews progress of cross-functional teams. Below it, **teamwork** is the operating mode: quality circles (Topic 4) at work-area level, cross-functional teams for problems that span departments (a shade problem spanning lab, dyeing and finishing cannot be circled by dyeing alone), and project teams for system changes. Teamwork under TQM is structured — charters, data, presentations, decisions recorded — not a hope that people will cooperate; and its meetings end with actions, owners and dates, like any other controlled process.

Glossary

QMS — quality management system. **ISO 9001** — certifiable QMS requirements standard. **Audit** — systematic independent examination against criteria, on evidence. **Nonconformity** — failure to fulfil a requirement; major vs minor by systemic seriousness. **QS 9000** — retired automotive QMS (successor IATF 16949). **EMS / ISO 14001** — environmental management system requirements. **Quality council** — senior body owning quality policy, review and resources.

Key points

• ISO 9001 = process approach + PDCA + seven principles; certification attests the system, surveillance keeps it honest.
• Audits: first-party (internal), second-party (customer), third-party (certification); value lies in the corrective-action loop, not the report.
• QS 9000 (→ IATF 16949) shows customer-specific systems layered on ISO 9001, directly relevant to automotive textile suppliers.
• ISO 14001 manages environmental aspects through the same PDCA logic; legal compliance is audited within the system.
• Quality councils own strategy and resources; circles and cross-functional teams do the improving.

Practice

1. A buyer's team audits a mill before placing orders, and an accredited body audits the same mill for ISO 9001 renewal. Classify both audits and state what each can and cannot conclude.
2. An internal audit finds the same calibration lapse in three departments. Major or minor nonconformity — and why?
3. Why is "we have ISO 9001, so this lot must be good" a logical error? Answer in the vocabulary of Topics 1 and 5.
4. Draft the agenda (five items) of a quarterly quality council for a composite mill, using this sheet's performance measures.

Answer key

1. The buyer's audit is **second-party**: it can conclude whether the mill meets *that buyer's* requirements and award or withhold business; it certifies nothing public. The accredited body's audit is **third-party**: it can conclude whether the QMS conforms to ISO 9001 and maintain or withdraw certification; it does not guarantee any individual lot and does not itself award orders.
2. **Major.** A lapse repeated across three departments is systemic — the calibration process itself is not effectively implemented — not an isolated slip. Majors require root-cause corrective action on the system (schedule control, responsibility, escalation on overdue items), not three local fixes.
3. Certification attests that a **system** conforming to ISO 9001 exists and is maintained (Topic 5). It raises the probability of conforming output but does not inspect or guarantee a specific lot; quality of the lot is **conformance to requirements** (Topic 1), demonstrable only by the lot's own inspection and test evidence. Confusing system assurance with product proof is exactly the QA-stage limitation the evolution in Topic 1 describes.

4. Model agenda: (1) Review of previous actions and audit status (internal, buyer, certification) including overdue corrective actions. (2) Quality performance: defect/DPMO trends, first-pass yield, cost of quality by category. (3) Customer feedback: complaints, claims, retention and buyer audit scores; supplier rating summary. (4) Improvement projects: status of circles and cross-functional teams against objectives; resource requests. (5) System changes: standard revisions, new buyer requirements, training plan — with decisions, owners and dates minuted. (Any five items tracing policy → data → action are acceptable.)

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib